

ISyE 6669 Midterm I

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Problem 1. Consider the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 0 & -1 \\ 1 & 3 & 1 \\ 4 & 2 & -1 \end{bmatrix}$. The rank of A is:

A 1

B 2

C 3

D 4

Solution: C

Problem 2. Consider the function $f(x, y) = x^3 + 2xy + 4y$. The first order Taylor's expansion of this function at $(1, 0)$ is

A $-2 + 6x + 3y$

B $-2 + 3x + 6y$

C $3 + x^2 + 3y$

D $3x^2 + 2y + 4$

Solution: B

Problem 3. The set $X = \{x \in \mathbb{R}^n \mid \sum_{i=1}^n x_i \leq 1\}$ is

A closed and bounded

B closed but not bounded

C not closed and not bounded

Solution: B

Problem 4. Which set is convex

- A $\{x \in \mathbb{R}^2 \mid e^{x_1} \geq x_2\}$
- B $\{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$
- C $\{x \in \mathbb{R}^2 \mid \max\{x_1, x_2\} \geq 1\}$
- D $\{x \in \mathbb{R}^2 \mid \min\{x_1, x_2\} \geq 1\}$

Solution: D

Problem 5. Which function is convex on the indicated domain

- A $f(x) = e^x$ on $[0, +\infty)$
- B $f(x) = x^3$ on $[-1, 1]$
- C $f(x) = -x^2 - 4x$ on $\mathbb{R}$
- D $f(x) = \sin(x)$ on $[0, 2\pi]$

Solution: A

Problem 6. Which statement is TRUE

- A Any optimization problem whose feasible region is unbounded cannot have an optimal solution.
- B Every global optimal solution to an optimization problem must have the same objective function value.
- C A convex optimization problem can have at most one global optimal solution.

Solution: B

Problem 7. Which of the following is a convex optimization problem

- A $\min \{x^2 + y^4 : x^2 + y^2 \geq 100\}$
- B $\max \{x^2 : x \in [-10, 10]\}$
- C $\min \{2x + 3y : 7x + y \leq 10, \ln(x) \geq 0\}$

Solution: C

Problem 8. What is the outcome of the problem $\max\{6x - y : x \geq 0, y \geq 0\}$

A Unbounded

B Optimal

C Infeasible

Solution: A

Problem 9. I have solved an optimization problem and got an optimal solution x^* . Suppose now a new constraint is added to my problem. If I find that x^* satisfies the new constraint, then x^* is an optimal solution of the modified problem.

A True

B False

Solution: A

Problem 10. If I maximize a univariate convex function over a nonempty closed and bounded interval, then there has to be an optimal solution which is one of the end points.

A True

B False

Solution: A

Problem 11. Consider an unconstrained optimization problem $\min f(x)$. Suppose at a point x_0 we know $\nabla f(x_0) = 0$ and $\nabla^2 f(x_0)$ is positive semidefinite. Then

A x_0 is a local minimizer

B x_0 is a local maximizer

C x_0 may be neither a local minimizer nor a local maximizer

Solution: C

Problem 12. The gradient vector of $f(x_1, x_2) = (x_1 - 2)^3 + 2x_1x_2 - 3x_2$ at point $(x_1 = 3, x_2 = 5)$ is

A $(12, 4)^\top$

B $(13, 3)^\top$

C $(14, 2)^\top$

D $(15, 1)^\top$

Solution: B

Problem 13. The Hessian matrix of $f(x_1, x_2) = (x_1 - 2)^3 + 2x_1x_2 - 3x_2$ at point $(x_1 = 3, x_2 = 5)$ is

A $\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$

B $\begin{bmatrix} 18 & 2 \\ 2 & 0 \end{bmatrix}$

C $\begin{bmatrix} 6 & 2 \\ 2 & 0 \end{bmatrix}$

D $\begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$

Solution: C

Problem 14. What is the derivative of $f(x) = |x|^3$ at $x = -2$

A 8

B -8

C 12

D -12

Solution: D

Problem 15. Consider the problem $\min \{c^\top x : x \in X\}$ where $\emptyset \neq X \subseteq \mathbb{R}^n$ is a full-dimensional compact set. Suppose $c \neq 0$. What can you say about the problem:

A The problem can be unbounded

B The problem can be infeasible

C The optimal solution is unique

D The optimal solution cannot be in the interior of X

Solution: D

Problem 16. Which of the following is not a feasible point of $X = \{x \in \mathbb{R}^2 \mid 2x_1 + x_2 \leq 4, -x_1 + 3x_2 \geq 3, x_1 \geq -2, x_2 \leq 3\}$

A $(-1, 2)$

B $(2, 0)$

C $(1, 2)$

D $(0, 3)$

Solution: B

Problem 17. For the transportation problem:

$$\begin{aligned} \min \quad & \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^m x_{ij} = d_j, \quad j = 1, \dots, n, \\ & \sum_{j=1}^n x_{ij} \leq s_i, \quad i = 1, \dots, m, \\ & x_{ij} \geq 0, \quad i = 1, \dots, m, \quad j = 1, \dots, n. \end{aligned}$$

Which of the following situations must cause infeasibility?

A The total supply $\sum_{i=1}^m s_i$ is strictly greater than the total demand $\sum_{j=1}^n d_j$.

B There exists a supply s_i which is greater than or equal to the total demand $\sum_{j=1}^n d_j$.

C The total supply $\sum_{i=1}^m s_i$ is strictly less than the total demand $\sum_{j=1}^n d_j$.

Solution: C

Problem 18. Choose correct statement(s) about the “here-and-now” decision and the “wait-and-see” decision.

A The “here-and-now” decision must be made before the uncertainty is realized.

B The “wait-and-see” decision can be made before part of the uncertainty is realized.

C The “here-and-now” decision can change after the uncertainty is realized.

Solution: A, B

Problem 19. Which of the following functions can be reformulated as convex piecewise linear functions?

A $f(x) = \sum_{k=1}^{2018} e^k |k \cdot x - \log(k)|, x \in \mathbb{R}.$

B $f(x_1, \dots, x_n) = \sum_{k=1}^n (-1)^k |x_k + 1|, (x_1, \dots, x_n) \in \mathbb{R}^n.$

C $f(x) = |x - 1| - |x - 2|, 0 \leq x \leq 2.$

Solution: A, C

Problem 20. Consider the following nonlinear optimization problem:

$$\begin{aligned} \min \quad & |x - 1| + 3|2 - x| \\ \text{s.t.} \quad & x \geq 0. \end{aligned}$$

Which of the following linear reformulations is/are equivalent to it?

A

$$\begin{aligned} \min \quad & w_1 + w_2 \\ \text{s.t.} \quad & w_1 \geq x - 1 \\ & -w_1 \leq x - 1 \\ & w_2 \geq 3(2 - x) \\ & -w_2 \leq 3(2 - x) \\ & x \geq 0 \end{aligned}$$

B

$$\begin{aligned} \min \quad & w_1 + 3w_2 \\ \text{s.t.} \quad & w_1 \geq x - 1 \\ & -w_1 \leq x - 1 \\ & w_2 \geq 2 - x \\ & -w_2 \leq 2 - x \\ & x \geq 0 \end{aligned}$$

C

$$\begin{aligned} \min \quad & w_1 + w_2 \\ \text{s.t.} \quad & w_1 \geq x - 1 \\ & -w_1 \leq x - 1 \\ & w_2 \geq 2 - x \\ & -w_2 \leq 2 - x \\ & x \geq 0 \end{aligned}$$

Solution: A, B

Problem 21. Consider the following nonlinear objective that averages over the ℓ_1 metric and ℓ_∞ metric

$$\min \quad \frac{1}{2}(|x+1| + |x-2|) + \frac{1}{2} \max\{|x-1|, |x-2|\}.$$

Which is the correct reformulation as a linear program?

A

$$\begin{aligned} \min \quad & \frac{1}{2}(z_1 + z_2) + \frac{1}{2}z_3 \\ \text{s.t.} \quad & -z_1 \leq x-1 \leq z_1 \\ & -z_2 \leq x-2 \leq z_2 \\ & -z_3 \leq x+1 \leq z_3 \\ & -z_3 \leq x-2 \leq z_3 \end{aligned}$$

B

$$\begin{aligned} \min \quad & \frac{1}{2}(z_1 + z_2) + \frac{1}{2}(x-1) + \frac{1}{2}(x-2) \\ \text{s.t.} \quad & -z_1 \leq x+1 \leq z_1 \\ & -z_2 \leq x-2 \leq z_2 \end{aligned}$$

C

$$\begin{aligned} \min \quad & \frac{1}{2}(z_1 + z_2) + \frac{1}{2}z_3 \\ \text{s.t.} \quad & -z_1 \leq x+1 \leq z_1 \\ & -z_2 \leq x-2 \leq z_2 \\ & -z_3 \leq x-1 \leq z_3 \\ & -z_3 \leq x-2 \leq z_3 \end{aligned}$$

Solution: C

Problem 22. Is the set $H = \{\mathbf{x} = (x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 : x_1 + x_2 - x_3 - x_4 = 0\}$ in $\mathbb{R}^5$ a polyhedron and why?

A No, because a polyhedron should be defined by inequalities.

B No, because a polyhedron has to be bounded, whereas a hyperplane is not bounded.

C Yes, because H is the intersection of two halfspaces, $\{\mathbf{x} \in \mathbb{R}^5 : x_1 + x_2 - x_3 - x_4 \leq 0\} \cap \{\mathbf{x} \in \mathbb{R}^5 : x_1 + x_2 - x_3 - x_4 \geq 0\}$.

Solution: C

Problem 23. Does a plane have any extreme point? Think about the plane $\{(x, y, z) \in \mathbb{R}^3 : x = 1\}$.

A Yes, any point on the plane is an extreme point.

B No, the plane does not have any extreme point, because any point on the plane is contained in a line which we know has no extreme point.

C The definition of extreme point does not apply to a plane, because a plane consists of lines.

Solution: B

Problem 24. Find the extreme ray(s) of the polyhedron $P = \{(x, y) \in \mathbb{R}^2 : x + y \leq 1, x \geq 0, y \geq 0\}$.

A $[1, 0]^\top$

B $[0, 1]^\top$

C The polyhedron P does not have any extreme ray.

Solution: C

Problem 25. Consider the polyhedron $P = \{(x, y, z) \in \mathbb{R}^3 : x - y \leq 1, x \geq 0, y \geq 0, z \geq 0\}$. Find all the correct representations of the point $[\frac{1}{2}, 2, 1]^\top$ as a convex combination of extreme points and conic combination of extreme rays of P .

A $[\frac{1}{2}, 2, 1]^\top = \frac{1}{2} \cdot [0, 0, 0]^\top + \frac{1}{2} \cdot [1, 0, 0]^\top + 2 \cdot [0, 1, 0]^\top + 1 \cdot [0, 0, 1]^\top$

B $[\frac{1}{2}, 2, 1]^\top = [0, 0, 1]^\top + \frac{3}{2} \cdot [0, 1, 0]^\top + \frac{1}{2} \cdot [1, 1, 0]^\top$

C $[\frac{1}{2}, 2, 1]^\top = \frac{3}{4} \cdot [0, 0, 0]^\top + \frac{1}{4} \cdot [1, 0, 0]^\top + \frac{7}{4} \cdot [0, 1, 0]^\top + \frac{1}{4} \cdot [1, 1, 0]^\top + 1 \cdot [0, 0, 1]^\top$

Solution: A, C

Problem 26. Consider the polyhedron $P = \{(x, y) \in \mathbb{R}^2 : x + y \leq 2, x \geq 0, y \geq 0\}$. Which of the following solutions is/are basic feasible solution(s)?

A $[0, 0]^\top$

B $[0, 1]^\top$

C $[1, 1]^\top$

D $[0, -1]^\top$

Solution: A

Problem 27. Given a polyhedron P in standard form where A is a 3×5 matrix with full row rank, how many basic feasible solutions can P possibly have?

A 0

B 1

C 10

D 100

Solution: A, B, C

Problem 28. Suppose a standard form LP has a basic feasible solution $\mathbf{x} = (x_1 = 1, x_2 = 2, x_3 = 0, x_4 = 3, x_5 = 0)$. Which of the following points can be a basic feasible solution adjacent to $\mathbf{x}$?

A $(2, 1, 3, 0, 0)$.

B $(1, 0, -1, 0, 0)$.

C $(0, 0, 0, 0, 3)$.

Solution: A, C

Problem 29. Suppose the direction to move in a simplex iteration is $\mathbf{d}^\top = (-2 \ -1 \ 0 \ 1 \ 0)$ and the current basic variables have values $(3, 1)$. Which variables are basic variables and what is the next iteration's basic feasible solution?

A Basic variables are (x_1, x_2) and the next iteration's basic feasible solution is $(1 \ 0 \ 0 \ 1 \ 0)^\top$.

B Basic variables are (x_3, x_5) and the next iteration's basic feasible solution is $(1 \ 0 \ 3 \ 0 \ 1)^\top$.

C Basic variables are (x_1, x_2) and the next iteration's basic feasible solution is $(0 \ 0.5 \ 0 \ 1.5 \ 0)^\top$.

Solution: A

Problem 30. If the reduced costs of nonbasic variables (x_3, x_4, x_5) are $(3, 0, -2)$ and the ratios are $(\frac{x_1}{-d_1}, \frac{x_2}{-d_2}) = (1, 0)$. Is this basic feasible solution degenerate and why?

- A Yes, because the basic variable x_2 must be 0.
- B No, because no basic variable can be zero.
- C Yes, because currently the reduced cost of x_4 is 0.

Solution: A

Problem 31. If the linear optimization problem

$$\begin{array}{ll} \min & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} = \mathbf{b} \end{array}$$

has two distinct feasible solutions $\mathbf{x}^1, \mathbf{x}^2$, then

- A the problem must be unbounded if $\mathbf{c}^\top \mathbf{x}^1 \neq \mathbf{c}^\top \mathbf{x}^2$.
- B if the optimal objective value is bounded then $\mathbf{c} = 0$.
- C either there are infinitely many optimal solutions or there is no optimal solution.

Solution: A, B, C